Building Materials

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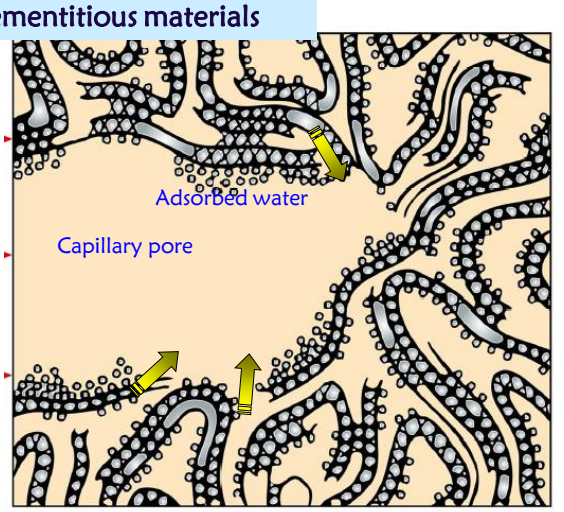
Chapter One	Introduction
Chapter Two	The Fundamentals
Chapter Three	Metal
Chapter Four	Concrete
Chapter Five	Asphalt
Chapter Six	Masonry
Chapter Seven	Wood
Chapter Eight	Composites

Chapter Two: The Fundamentals of Materials

1. Physical Architecture of Materials
2. Behavior of Materials Under Stress
3. Surface Properties
4. Strength and Failure

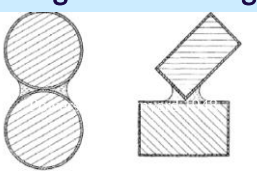
2.3 Surface Properties

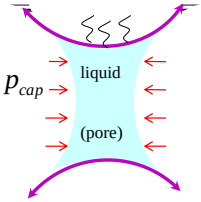
Water in cementitious materials

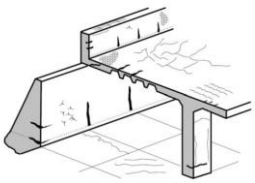


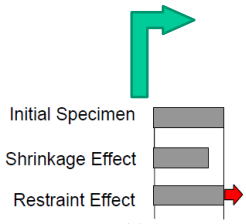
2.3 Surface Properties

Shrinkage and Cracking



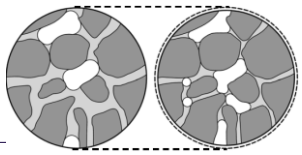






$$d\varepsilon(t,\xi) = \frac{d\sigma(\xi)}{E_\sigma(\xi)} + d\varepsilon_{SHR}(\xi)$$

Can shrinkage causes cracking?

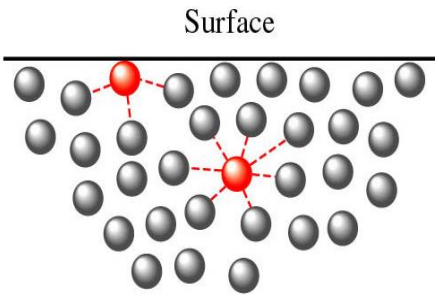


2.3 Surface Properties

Surface Tension of a Liquid

- The resistance of a liquid to an increase in its surface area is called the SURFACE TENSION of a liquid.

The properties of a surface are quite different from the properties of the bulk material.



2.3 Surface Properties

Surface Tension of a Liquid

Effects

- Liquids tend to minimum surface area, droplets form spherical shape.
- Stronger intermolecular forces result in greater surface tension (Hg > H_2O).
- A substance with a low surface tension will evaporate readily.

2.3 Surface Properties

Surface Tension of a Liquid

Surface tension of common liquid

Liquid	Surface Tension γ (N/m)
Benzene (20 °C)	0.029
Blood (37 °C)	0.058
Glycerin (20 °C)	0.063
Mercury (20 °C)	0.47
Water (20 °C)	0.073
Water (100 °C)	0.059

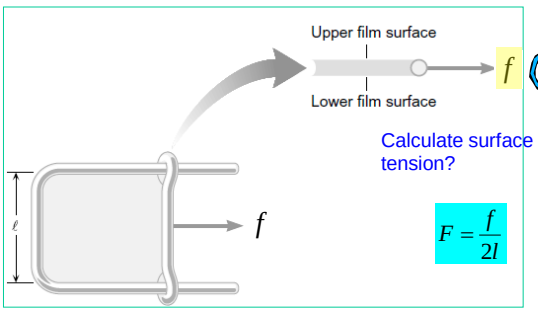
2.3 Surface Properties

Surface Tension of a Liquid

The surface tension T is the magnitude F of the force exerted parallel to the surface of a liquid divided by the length L of the line over which the force acts:

$$T = \frac{F}{L}$$

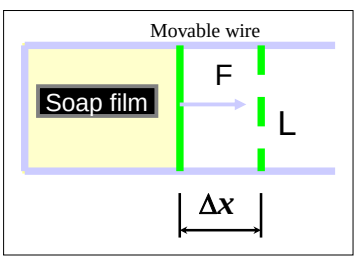
Unit of surface tension: N/m



2.3 Surface Properties

Surface Tension of a Liquid

surface energy vs. surface tension.



T is surface tension (N/m)

γ_s : is surface energy (J/m^2)

Energy input by force required to move wire stationary:

$$2T \cdot L \cdot \Delta x$$

Output of the work as an increase in surface energy :

$$\gamma_s \cdot 2 \cdot L \cdot \Delta x$$

$$\Downarrow$$

$$T = \gamma_s$$

For liquid: surface energy = surface tension.

2.3 Surface Properties

Surface Tension of a Solid

Solid…

work $W = 2TL\Delta x + W_d$ — Deformation energy

Surface energy $\gamma = \frac{W}{\Delta A} = \frac{W}{2L\Delta x} = T + \frac{W_d}{2L\Delta x} > T$

For solid: surface energy > surface tension.

2.3 Surface Properties

Surface Tension of a Liquid

Liquid	Surface Tension γ(N/m)
Benzene (20 °C)	0.029
Blood (37 °C)	0.058
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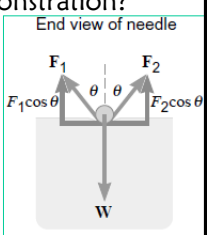
EXAMPLE: Floating a needle on the surface of water

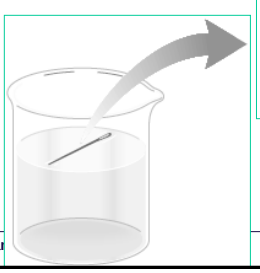
A needle has a length of 3.2cm. When placed gently on the surface of the water in a glass, this needle will float if it is not too heavy. What is the weight of the heaviest needle that can be used in this demonstration?

Solution: $\gamma_s = 0.073N/m$

$$W = 2(\gamma_s L)\cos\theta$$

$$W = 2(\gamma_s L)\cos\theta$$
$$= 2(0.073N/m)(0.032m)\cos 0^0$$
$$= 4.7\times10^{-3}N$$

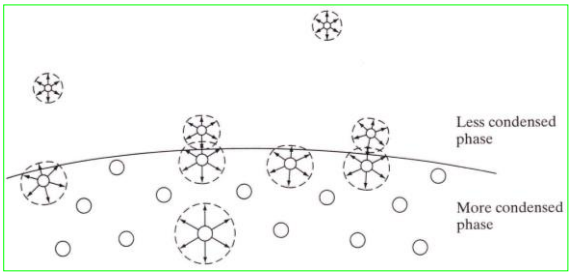




2.3 Surface Properties

Interface

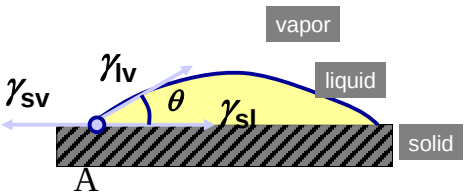
- The interfacial energy depends on the attractions across the interface.
- The existence of attractions between molecules across the interface leads to three important phenomena: wetting, adsorption, and capillary effects.



2.3 Surface Properties

Wetting

- A liquid is wetting when the attractions between its atoms and those of the surface atoms are greater than between the atoms in the liquid.



Force balance: $\gamma_{LV}\cos\theta+\gamma_{SL}=\gamma_{SV}$

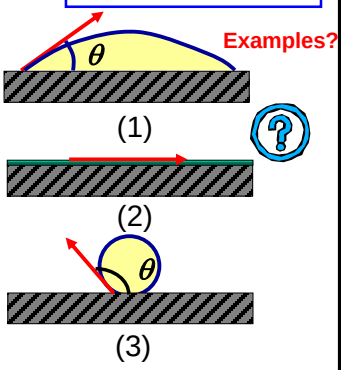
2.3 Surface Properties

Wetting

Wetting angle $\cos\theta = \dfrac{\gamma_{sv} - \gamma_{sl}}{\gamma_{lv}}$

(1) Water on glass
(2) Water on wood
(3) Mercury on concrete

Examples?



（1）$0<\gamma_{sv}-\gamma_{sl}<\gamma_{lv}$, then $\theta<90^o$, partial wetting occur.

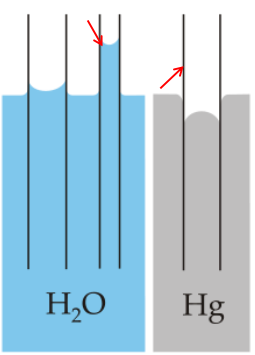
（2）$\gamma_{sv}-\gamma_{sl}=\gamma_{lv}$, then $\theta=0$，wetting occur

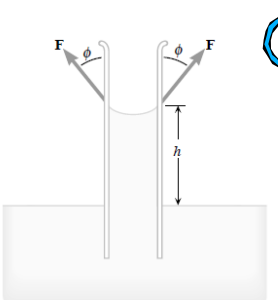
（3）$\gamma_{sv}<\gamma_{sl}$, then $\theta>90^o$, nonwetting occur

2.3 Surface Properties

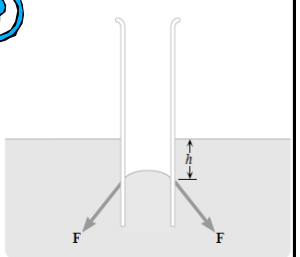
Capillary Effect

Forces, which are between like molecules, are called cohesive forces.
Force, which are between unlike molecules, are called adhesive forces.





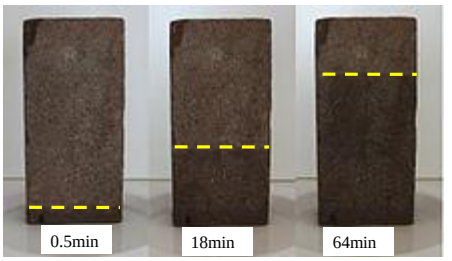
the adhesive forces are stronger than the cohesive forces



adhesive forces are weaker than the cohesive forces

2.3 Surface Properties

Capillary Effect



Capillary flow in a brick



2.3 Surface Properties

Capillary Effect

Liquid is drawn up into the capillary by a circumferential surface force:

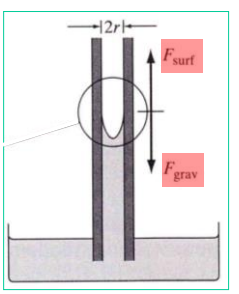
$$F_{surf} = 2\pi r \gamma_{LV} \cdot \cos\theta$$

Gravitational force:

$$F_{grav} = \pi r^2 h g \rho$$

At equilibrium:

$$h = \frac{2\gamma_{LV} \cdot \cos\theta}{rg\rho}$$



In the case of water in concrete, h is about 1-10m

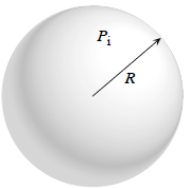
If gravitational forces are not important, then at equilibrium, a pressure will be exerted on the water:

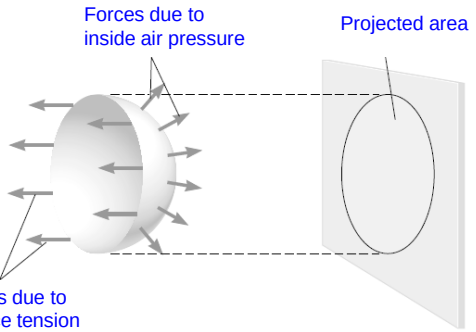
Laplace equation: $P = \frac{2\gamma_{LV} \cdot \cos\theta}{r}$

2.3 Surface Properties

Capillary Effect

Pressure inside a soap bubble and a liquid drop





$$-2\gamma(2\pi R)+P_i(\pi R^2)=0$$

A spherical soap bubble: $P_i=\frac{4\gamma}{R}$

A liquid drop: $P_i=\frac{2\gamma}{R}$

2.3 Surface Properties

Capillary Effect

A liquid drop: $P_i=\frac{2\gamma}{R}$ **Laplace equation**

Q: Using the Laplace Equation, compare $\Delta P=P_i-P_o$ for ideal spherical bubbles of 1 mm and 10 nm in water at ambient pressure?

A spherical soap bubble: $P_i-P_o=\frac{4\gamma}{R}$

$$\Delta P_{1mm}=\frac{2\cdot\gamma_{H_2O}}{r}=\frac{2\cdot\left(0.0728\ \mathrm{N\cdot m^{-1}}\right)}{0.5x10^{-3}\ \mathrm{m}}=\mathbf{291\ Pa}$$

$$\Delta P_{10nm}=\frac{2\cdot\left(0.0728\ \mathrm{N\cdot m^{-1}}\right)}{0.5x10^{-9}\ \mathrm{m}}=\mathbf{2.91x10^{8}\ Pa}$$

A liquid drop: $P_i-P_o=\frac{2\gamma}{R}$

2.3 Surface Properties

Kelvin equation

The Effects of Curved Surfaces

- Highly curved surfaces (small r) have reduced coordination of surface atoms and are inherently unstable relative to a surface with low curvature.

- This condition leads to enhanced evaporation and therefore, higher vapor pressure compared to a bulk planar surface.

2.3 Surface Properties

Kelvin equation

The Kelvin equation describes the change in vapor pressure due to a curved liquid/vapor interface:

$$\ln\left(\frac{P}{P_0}\right) = -\frac{2\gamma V_m}{RTr}$$

$$\ln(RH) = \ln\left(\frac{P}{P_0}\right)$$

where p is the actual vapor pressure, p_0 is the saturated vapor pressure, γ is the surface tension, V_m is the molar volume, R is the universal gas constant, r is the radius, and T is temperature in K ($273+^0C$).

For water vapor: $V_m = 18 \times 10^{-6} m^3/mol$

2.3 Surface Properties

Kelvin equation

$$\ln\left(\frac{P}{P_0}\right) = -\frac{2\gamma V_m}{RTr}$$

↓

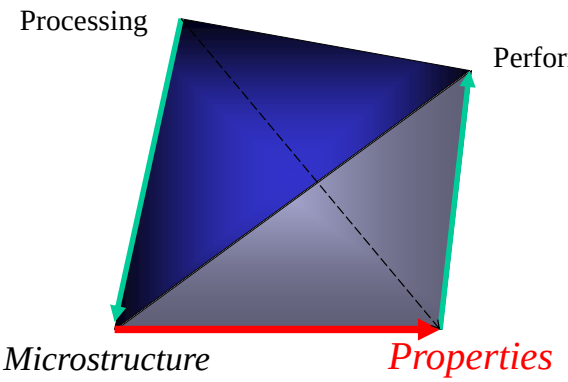
Laplace equation: **(for meniscus)**

$$P = -\frac{RT}{V_m}\ln\left(\frac{p}{p_0}\right) = \frac{2\gamma}{r}$$

Chapter Two: The Fundamentals of Materials

1. Physical Architecture of Materials
2. Behavior of Materials Under Stress
3. Surface Properties
4. Strength and Failure

2.4 Strength and Failure



2.4 Strength and Failure



Alaska MD-80 crash (1999)



Death 88 people

Excessive wear on stabilizer jackscrew



Challenger (1986)



failure of an O-ring seal (polymer)

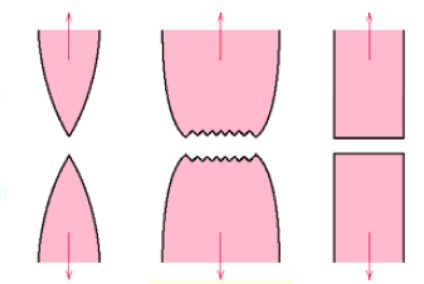
2.4 Strength and Failure

So, failure sometimes takes place!!!
Why does it occur?
What actually happens during fracture?
How can we prevent it?

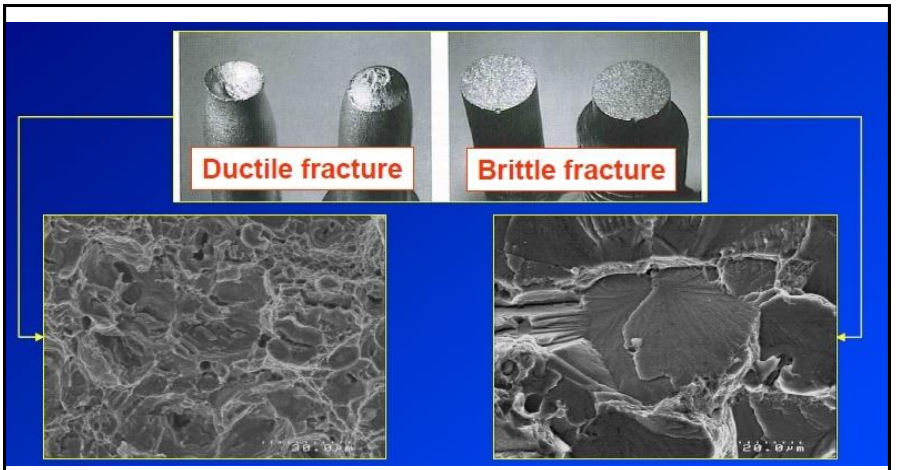
2.4 Strength and Failure

Types of Failure

- Most fundamental types of failure are ductile and brittle.
- Fracture involves crack initiation and propagation under an applied stress.
- Cracks may exist but they are stable and do not propagate under the particular loading conditions.





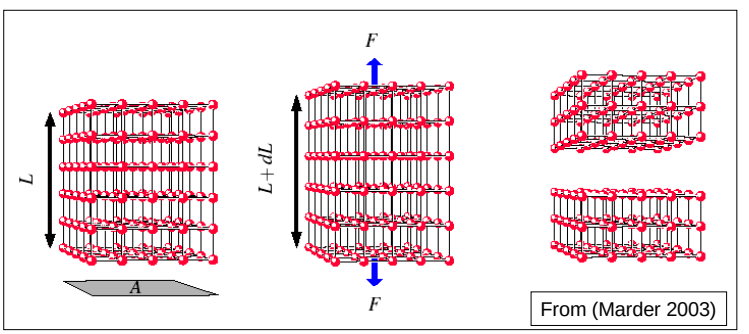


High energy is absorbed during ductile failure (high energy fracture mode)
Less catastrophic

Low energy is absorbed during brittle fracture(low energy fracture mode)
More catastrophic

2.4 Strength and Failure

Theoretical cohesive strength

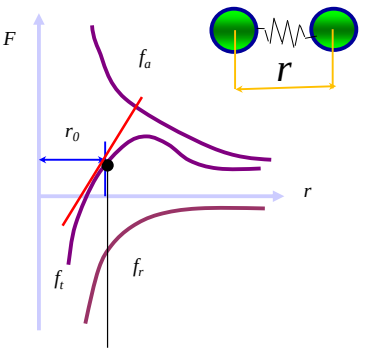


From (Marder 2003)

2.4 Strength and Failure

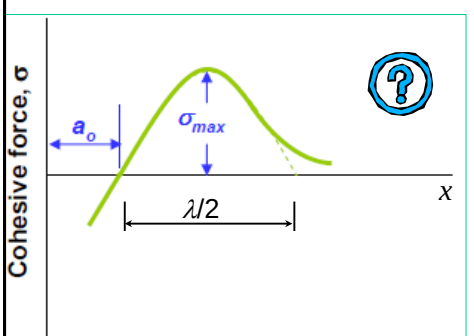
Theoretical cohesive strength

- In the most basic term, strength is due to the cohesive forces between atoms.
- The attractive and repulsive force acting on the two atoms lead to cohesive force between two atoms which varies in relation to the separation between these atoms.



2.4 Strength and Failure

Theoretical cohesive strength



The theoretical cohesive strength σ_{max} can be obtained in relation to the sine curve:

$$\sigma = \sigma_{th} \sin \frac{2\pi x}{\lambda}$$

$\sigma_{th} = \sigma_{max}$: is the theoretical strength

λ : is wave length of sine curve

a_0 : is the unstrained interatomic spacing or r_0

work required to separate the surface

$$W = \int_0^{\lambda/2} \sigma_{th} \sin \frac{2\pi x}{\lambda} dx = \frac{\lambda \sigma_{th}}{\pi}$$

2.4 Strength and Failure

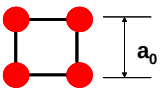
Theoretical cohesive strength

r : is surface energy

$$W = \frac{\lambda \sigma_{th}}{\pi} = 2\gamma \quad \Rightarrow \quad \sigma_{th} = \frac{2\pi\gamma}{\lambda} \qquad (1)$$

When deformation is small: $\sigma = \sigma_{th} \sin\frac{2\pi x}{\lambda} \approx \sigma_{th}\frac{2\pi x}{\lambda}$

Hooke's law $\sigma = = E\varepsilon = E\frac{x}{a_0}$



$$\lambda = \frac{\sigma_{th} 2\pi a_0}{E} \qquad (2)$$

From (1) and (2), we get:

$$\sigma_{th} = \sqrt{\frac{E\gamma}{a_0}}$$

In general:

$$\gamma \approx \frac{a_0 E}{100} \quad \Rightarrow \quad \sigma_{th} = \frac{E}{10}$$

2.4 Strength and Failure

Theoretical cohesive strength

Material	Young's Modulus GPa	Theoretical Strength GPa	Practical Strength GPa	Practical/Theoretical
Iron	195-205	43-56	0.3	0.006
Copper	110-130	24-55	0.2	0.005
Titanium	110	31	0.3	0.009
Silicon	110-160	45	0.7	0.01
Glass	70	37	0.4	0.01
Plexiglas	3.6	3	0.05	0.01



- This theoretical cohesive strength is exceptionally higher than the practical strength of engineering materials.
- This difference between cohesive and fracture strength is due to inherent flaws or defects in the materials which lower the fracture strength in engineering materials.

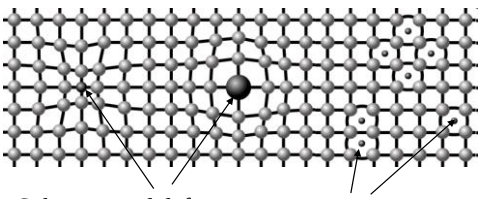
2.4 Strength and Failure

Theoretical cohesive strength

Observed fracture strength is always lower than theoretical cohesive strength

Griffith explained that the discrepancy is due to the ***inherent defects*** in brittle materials leading to stress concentration. → lower the fracture strength of the materials

Imperfections leading to strength properties





2.4 Strength and Failure

Fracture stress for realistic material

Inglis (1913) analyzed for the flat plate with an elliptical hole with major axis 2a and minor axis 2b, subjected to far end stress σ, the stress at the tip of the major axis (point A) is given by:

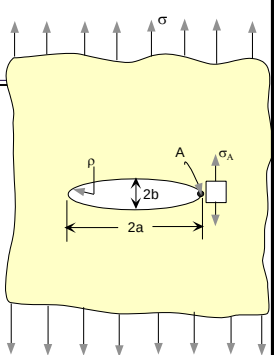
$$\sigma_A = \sigma\left(1+\frac{2a}{b}\right)$$

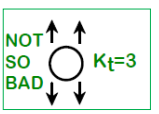
The ratio σ_A/σ is defined as the stress concentration factor, k_t

When a = b, it is a circular hole, then $k_t = 3$.

When b is very very small, Inglis define radius of curvature as $\rho = \frac{b^2}{a}$.

And the tip stress is: $\sigma_A = \sigma\left[1+2\left(\frac{a}{\rho}\right)^{1/2}\right]$





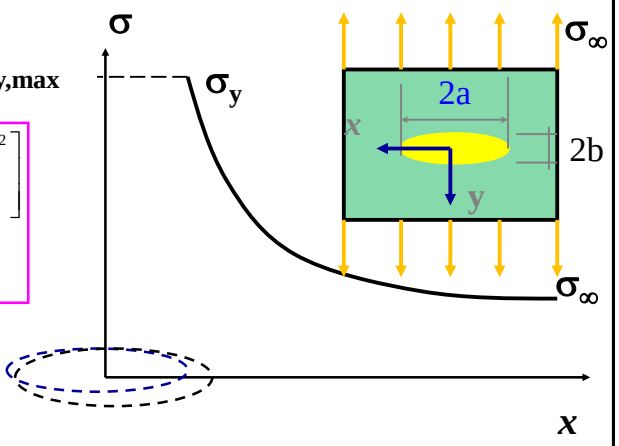


2.4 Strength and Failure

Stress Concentration

$$\sigma_{y\max} = \sigma_\infty \left[1 + 2\left(\frac{a}{\rho}\right)^{1/2}\right]$$

$$\rho = \frac{b^2}{a}$$



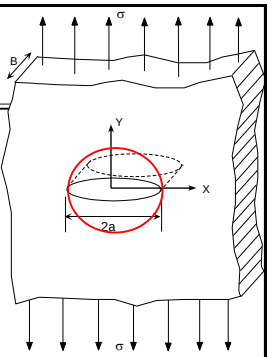
2.4 Strength and Failure

Griffith Theory of Brittleness

Griffith proposed that:

- There is a population of defects in all brittle materials of various sizes, shapes and orientations. When an applied stress creates a concentrated stress at one of these defects which is higher than the theoretical strength then the defect of crack propagates and fracture occurs.
- during crack propagation elastic strain energy is released.
- fracture creates new surfaces which raises the surface energy.
- derived a critical stress for propagation of an elliptical crack.

2.4 Strength and Failure



Griffith Theory of Brittleness

Energy released due to crack with length of 2a:

$$U = \frac{1}{2}\sigma\varepsilon V = \frac{1}{2}\frac{\sigma^2}{E}\left(\frac{4}{3}\pi a^3\right)$$

Strain energy release rate (per unit area):

$$G = \left|\frac{\partial U}{\partial A}\right| = \frac{1}{2\pi a}\left|\frac{\partial U}{\partial a}\right| = \frac{a\sigma^2}{E}$$

Released energy is used to create fracture surfaces:

$$G = G_c = 2\gamma \Rightarrow \frac{a\sigma_c^{\ 2}}{E} = 2\gamma$$

Fracture strength:

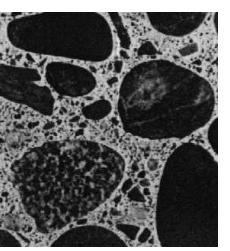
$$\sigma_c = \sqrt{\frac{2E\gamma}{a}}$$

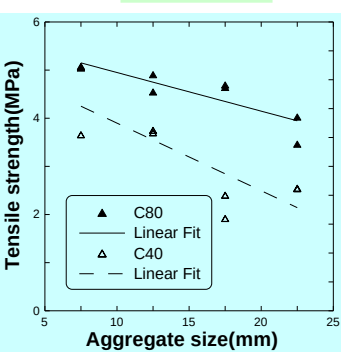
When $a \to 0$, $\sigma_c \to \infty$

2.4 Strength and Failure

Griffith Theory of Brittleness

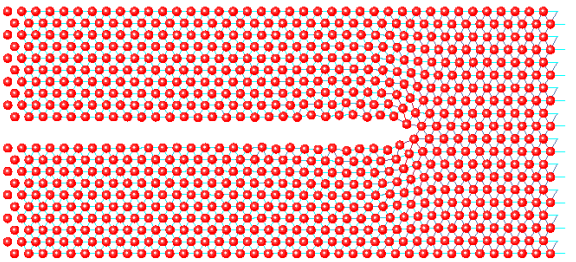
$$\sigma_c = \sqrt{\frac{2E\gamma}{a}} \Rightarrow \sigma_c = k\sqrt{\frac{2E\gamma}{R_{max}}}$$





2.4 Strength and Failure

Griffith Theory of Brittleness



$$\sigma_c = \sqrt{\frac{2E\gamma}{a}} \quad \Rightarrow \quad \sigma_c = \sqrt{\frac{2E(\gamma + \gamma_P)}{a_c}}$$

Summary

3. Surface Properties

Surface tension

Wetting (interface)

Capillary effect (interface)

4. Failure and Fracture

Theoretical strength

Griffith theory of brittleness



QUESTIONS?